

DEPARTMENT OF CHEMISTRY & CHEMICAL ENGINEERING

CHM-302: Organic Synthesis & Reaction Kinetics

Experiment 04: Catalyzed Esterification & Recrystallization

Term: Fall 2026
Duration: 3.5 Hours
Lab Location: Hall 402
Total Points: 50 Pts

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Instructor: Dr. Marcus Vance

Head TA: Elena Rostova

Pre-Lab Due: Oct 28, 2026

Format: Pair Work

Student 1 Name: _____

Student ID: _____

Student 2 Name: _____

Bench #: _____

1 Executive Summary & Learning Objectives

This laboratory assignment focuses on the acid-catalyzed esterification of salicylic acid with acetic anhydride to yield acetylsalicylic acid (aspirin), followed by crude isolation, purificative recrystallization in an ethanol-water solvent system, and quantitative assessment of chemical purity via micro-melting point determination and ferric chloride (Fe^{3+}) complexation spectroscopy.

Key Pedagogical Objectives:

1. Perform nucleophilic acyl substitution under anhydrous conditions using concentrated acid catalysis.
2. Master vacuum filtration techniques and compute accurate theoretical, crude, and purified percent yields.
3. Understand recrystallization mechanics based on temperature-dependent solubility differentials.
4. Assess organic product purity using colorimetric complexation assays and melting point range analysis.

2 Chemical Hazards & Laboratory Safety Protocols

MANDATORY SAFETY DIRECTIVES & GHS HAZARDS

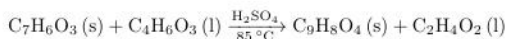
All operations involving acetic anhydride and concentrated sulfuric acid MUST be conducted inside operational fume hoods. Wear splash goggles, nitrile gloves, and lab coat at all times.

Reagent	CAS #	Primary GHS Hazard Statements	PPE Requirement
Salicylic Acid	69-72-7	Acute Tox. 4 (Oral), Eye Dam. 1 (H318, H302)	Nitrile Gloves, Safety Glasses
Acetic Anhydride	108-24-7	Flam. Liq. 3, Skin Corr. 1B, Acute Tox. 4 (H226, H314)	Fume Hood, Heavy Nitrile
Sulfuric Acid (18 M)	7664-93-9	Skin Corr. 1A, Met. Corr. 1 (H314, H290)	Double Gloves, Face Shield
Acetylsalicylic Acid	50-78-2	Acute Tox. 4, Skin Irrit. 2 (H302, H315)	Standard Lab Coat, Goggles

Emergency Procedure: Flush chemical skin exposure with water for 15+ minutes. Benchtop acid spills must be neutralized with sodium bicarbonate (NaHCO_3) prior to wipe-down. Waste solutions containing heavy metals (FeCl_3) must be poured into Hazardous Metal Waste Container B.

3 Theoretical Reaction Scheme & Stoichiometric Matrix

Reaction Scheme: Acid-Catalyzed Acetylation of Salicylic Acid



Reagent Physical Property Table

Chemical Compound	MW (g/mol)	Density (g/mL)	m.p. ($^\circ\text{C}$)	b.p. ($^\circ\text{C}$)	Reaction Role
Salicylic Acid	138.12	1.443	158.6	211.0	Limiting Reagent
Acetic Anhydride	102.09	1.082	-73.1	139.8	Excess Reactant / Solvent
Sulfuric Acid (98%)	98.08	1.840	10.3	337.0	Acid Catalyst
Acetylsalicylic Acid	180.16	1.400	135.0-136.0	Decomposes	Target Product
Water (Deionized)	18.02	1.000	0.0	100.0	Quenching Solvent